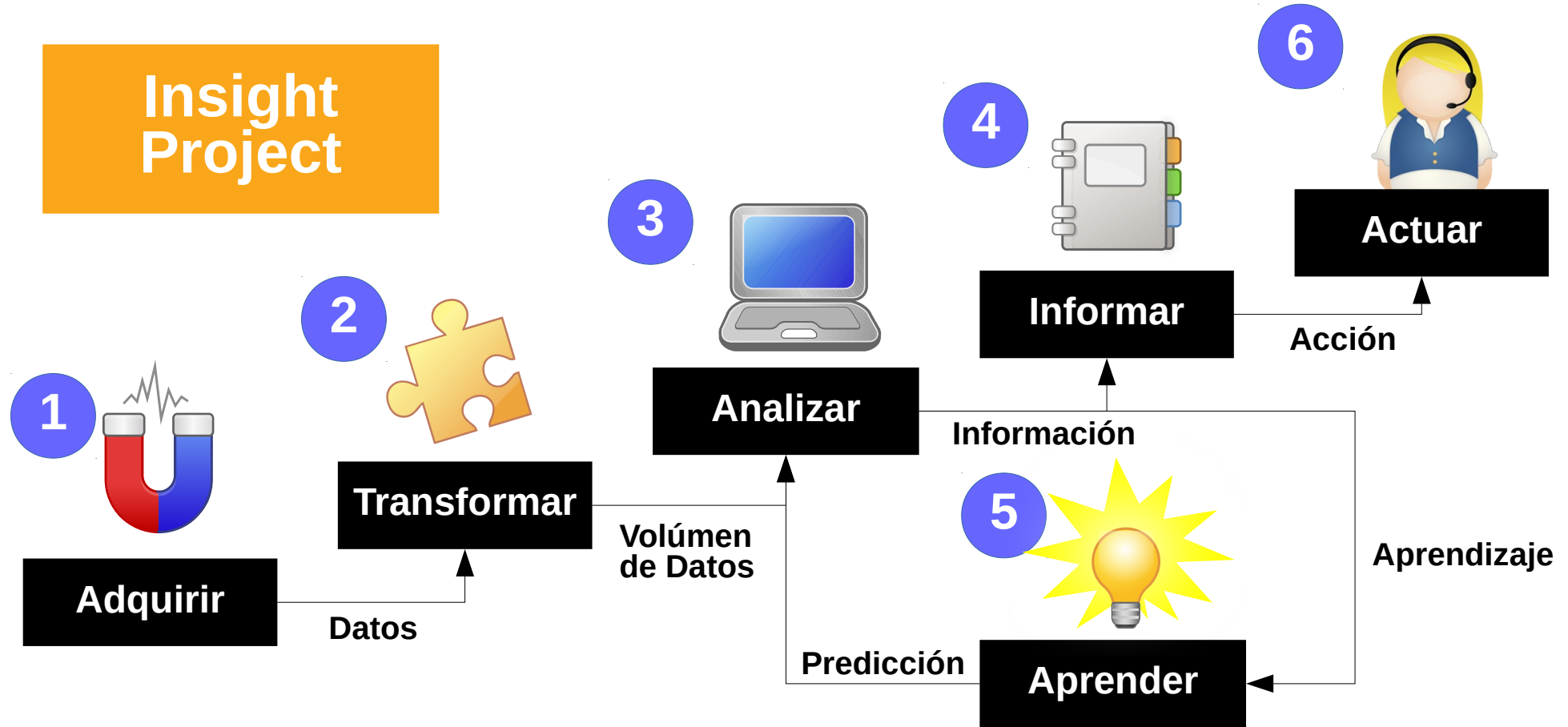


# Intelligent Monitoring & Analytics

## Insight Project



# Monitoring Silos

**NPMD**

**Network Performance Monitoring and Diagnostics**

**ITM**

**Information Technology Monitoring**

**APM**

**Application Performance Monitoring**

**DEM**

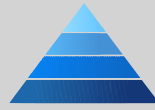
**Digital Experience Monitoring:** “Real time user monitoring and analytics over Web and mobile application”

# Insight :: Layers (4)

## 1. Visualization



Terminal



Portal



Grapher



Guardian

## 2. Analysis



Querier



Analyzer



Actioner



Notifier



Configurer

## 3. Storage



TS  
DB



SQL  
DB



LDAP  
DB

Persistent  
Cache

In- Memory  
Queues

## 4. Collection

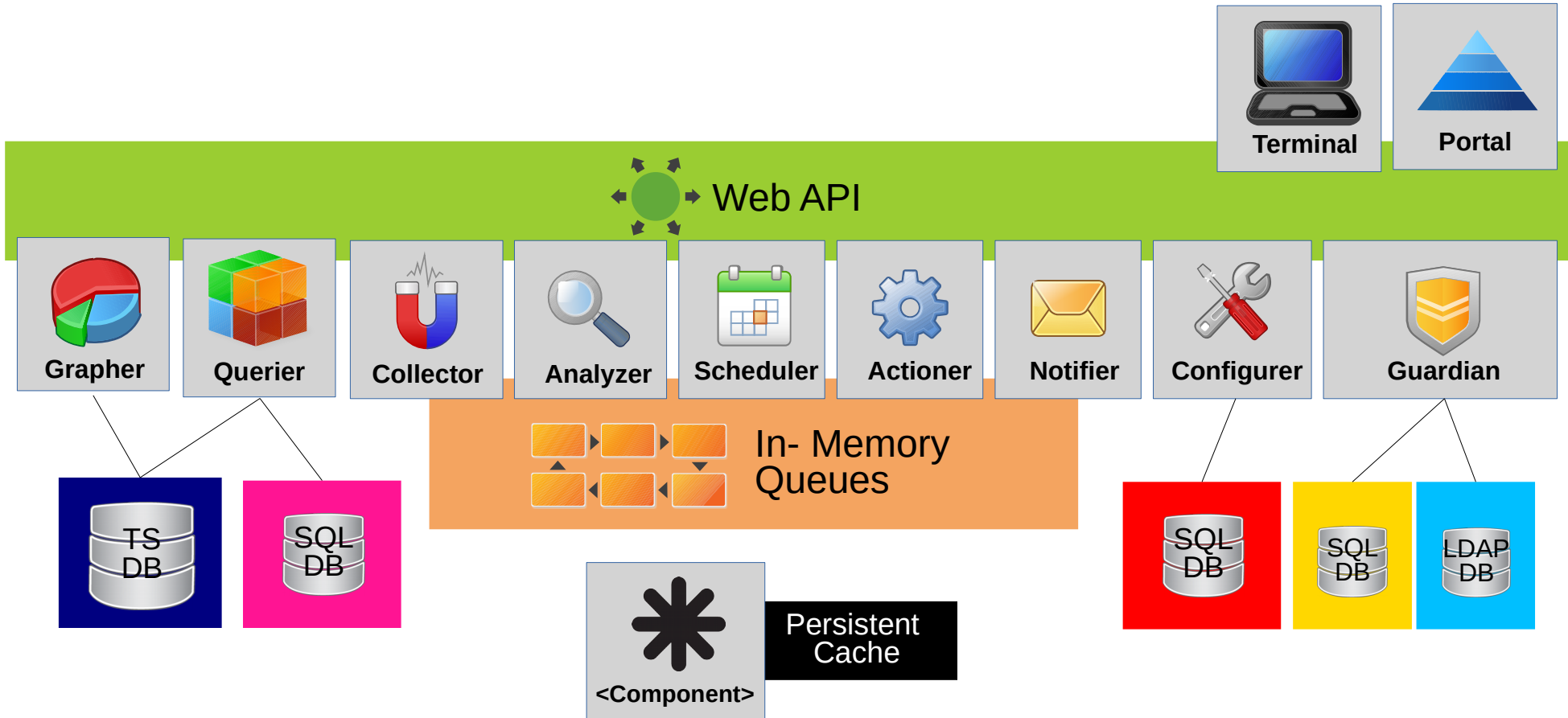


Collector

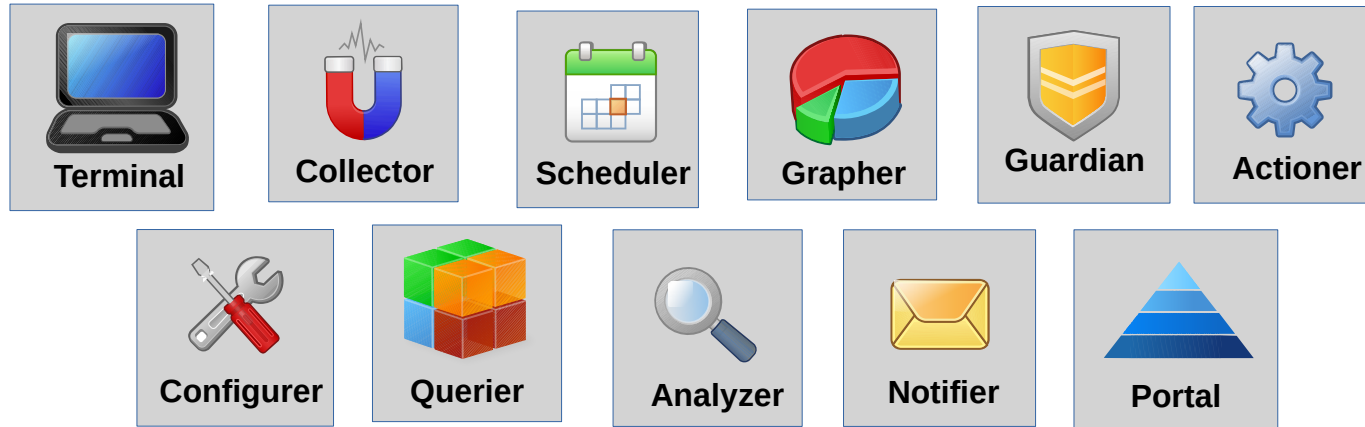


Scheduler

# Insight :: Components (18)



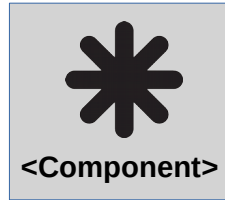
# Insight :: Components



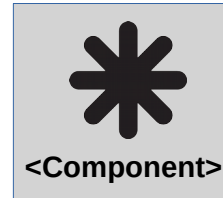
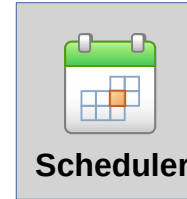
**Roadmap**

# Insight :: Components

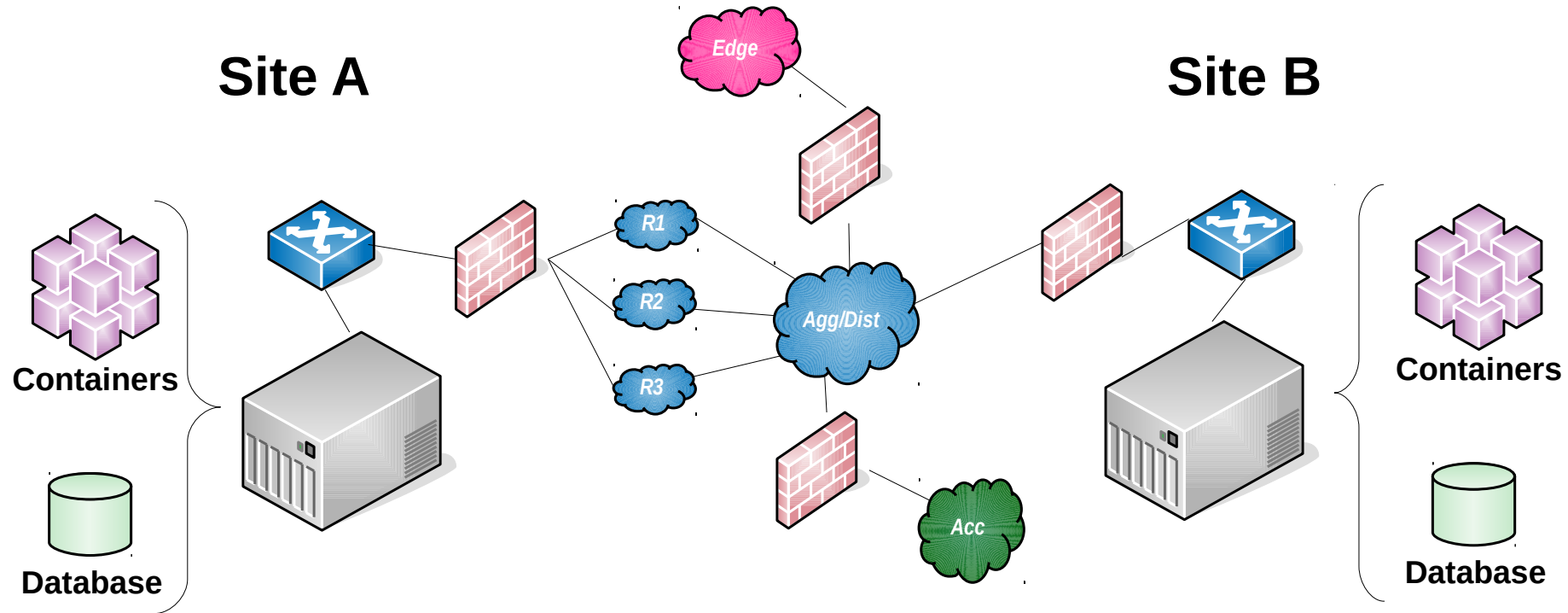
**On  
Cloud**



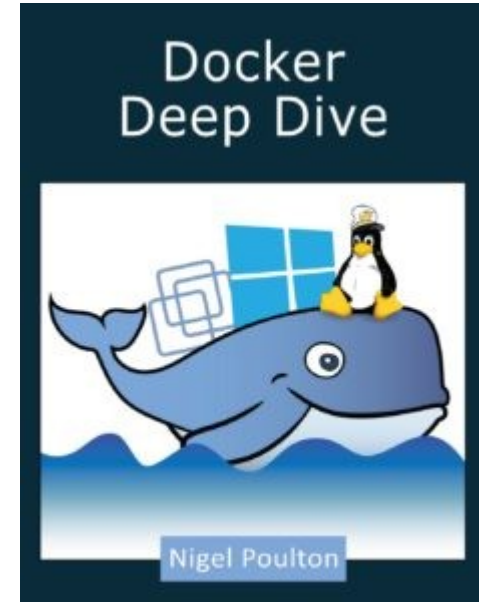
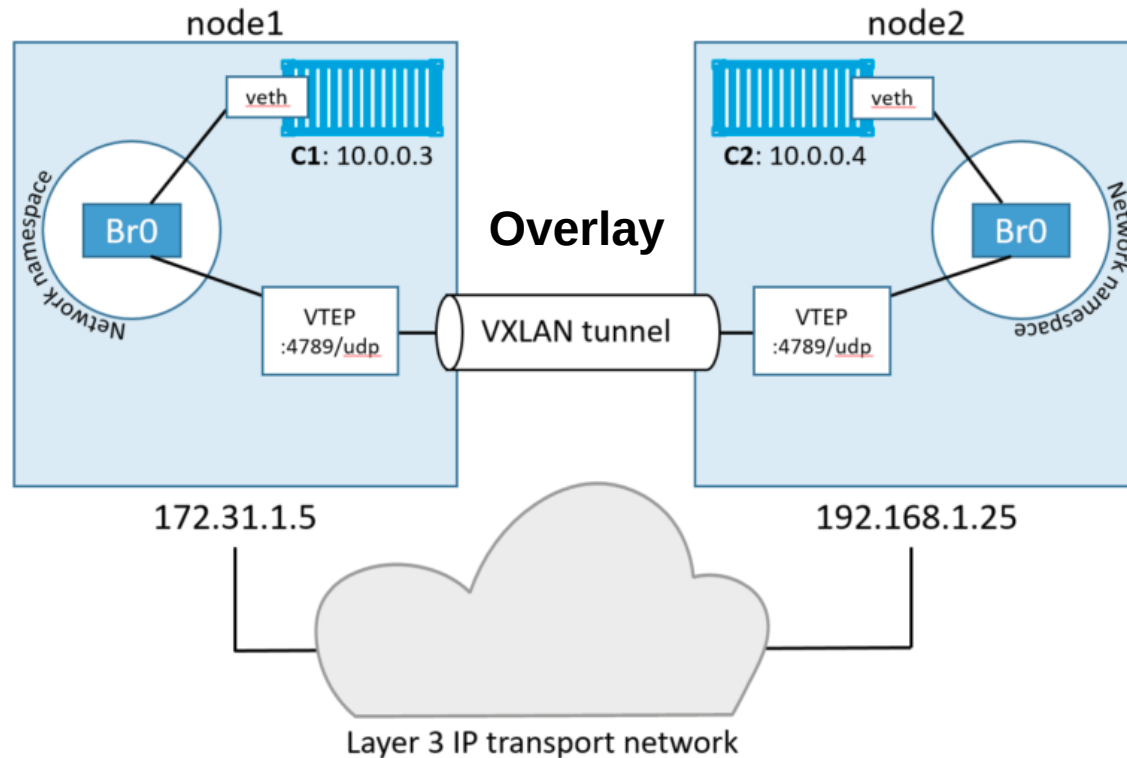
**On “Any”  
Premise**



# Diagrama de Redes (Heterogénea)



# Diagrama de Redes (Homogénea)



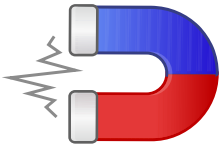
<https://success.docker.com/article/networking>  
<https://blog.nigelpoulton.com/demystifying-docker-overlay-networking/>



# Insight :: Components



**Configurers:** mantienen en tiempo real la unicidad y sincronía de la configuración global (compartida) y específica (individual) de los componentes del sistema.



**Collectors:** recolectan de forma continua los datos a través de la interacción directa con las interfaces de comunicación de los objetos (fuentes de datos) de la ciber-infraestructura.



**Schedulers:** planifican y distribuyen la recolección activa de datos de la ciber-infraestructura.

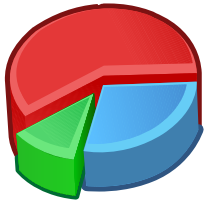
# Insight :: Components



**Analyzers:** inspeccionan los datos recolectados e identifican eventos, comportamientos, tendencias y causa raíz sobre situaciones de riesgo asociadas a los objetos monitorizados de la ciber-infraestructura.



**Queriers:** ejecutan consultas hacia las bases de datos y las devuelven en formato único legible por todos los componentes.



**Graphers:** presentan gráficas estadísticas sobre la información recolectada de los objetos de la ciber-infraestructura.



**Guardians:** mantienen las reglas de autenticación, autorización y acceso de los usuarios registrados, basada en roles para centralizar y controlar el uso de los componentes del sistema.

# Insight :: Components



**Actioners:** ejecutan acciones puntuales definidas por los usuarios o periódicas de acuerdo con instrucciones de los analizadores.



**Notifiers:** gestiona el envío de notificaciones a través de interfaces de comunicación con sistemas y personas.



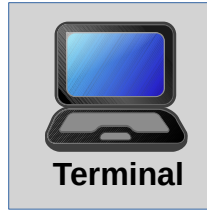
**Portal:** presenta una interfaz gráfica Web que permite interactuar con los componentes del sistema (Usuario Final / Administrador).



**Terminal:** presenta una interfaz gráfica textual que permite interactuar con los componentes del sistema (Administrador).

# Insight :: Component Stack

1



2

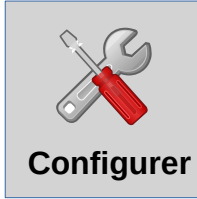


3

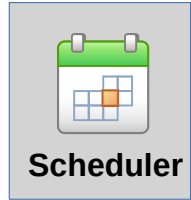


# Insight :: Component Stack

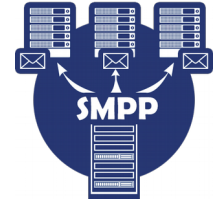
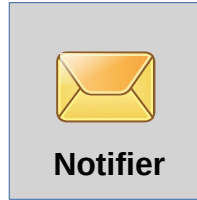
4



5



6

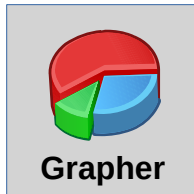


7



# Insight :: Component Stack

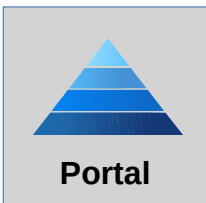
8



9



10



11



# Insight :: Database Stack

12



ubuntu



docker



*influxdb*

13



ubuntu



docker



redis



RabbitMQ

14



ubuntu



docker



15



ubuntu



docker

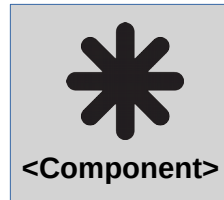


# Insight :: Database Stack

16



17

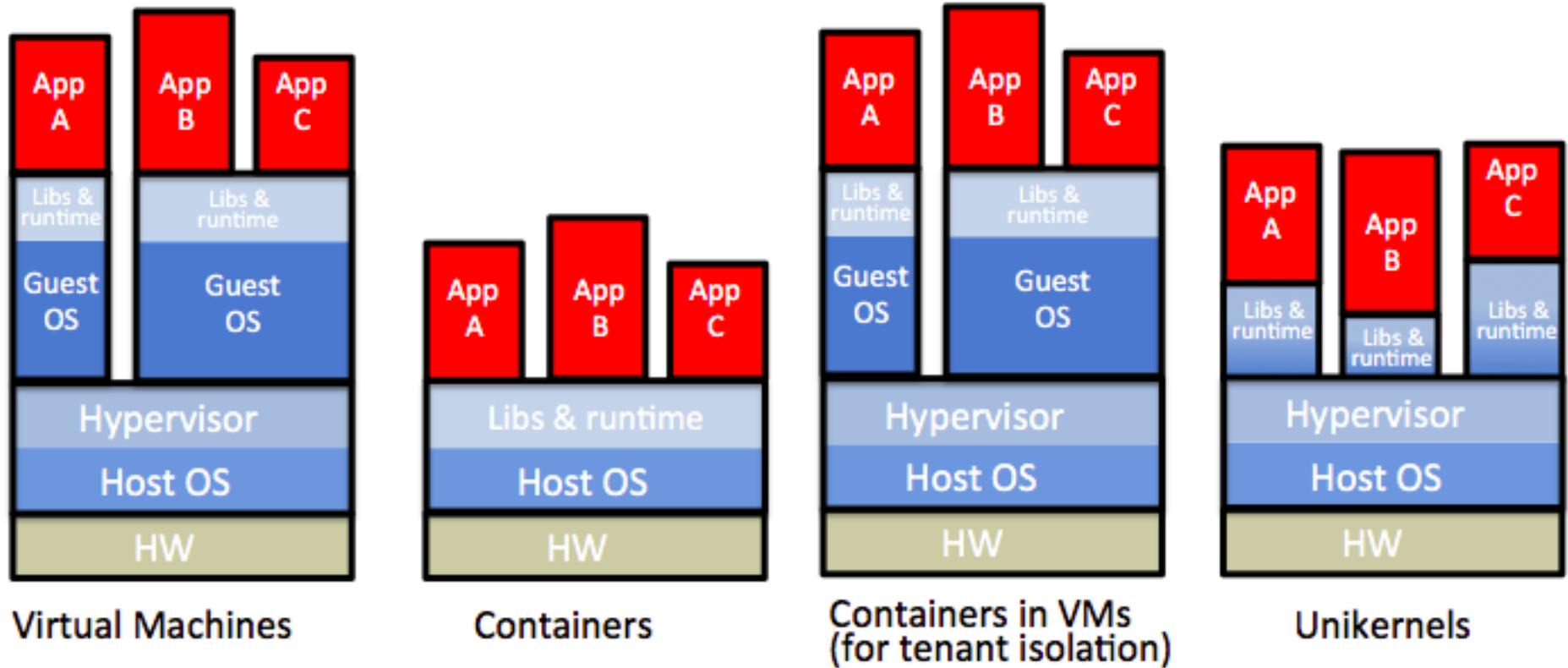


18

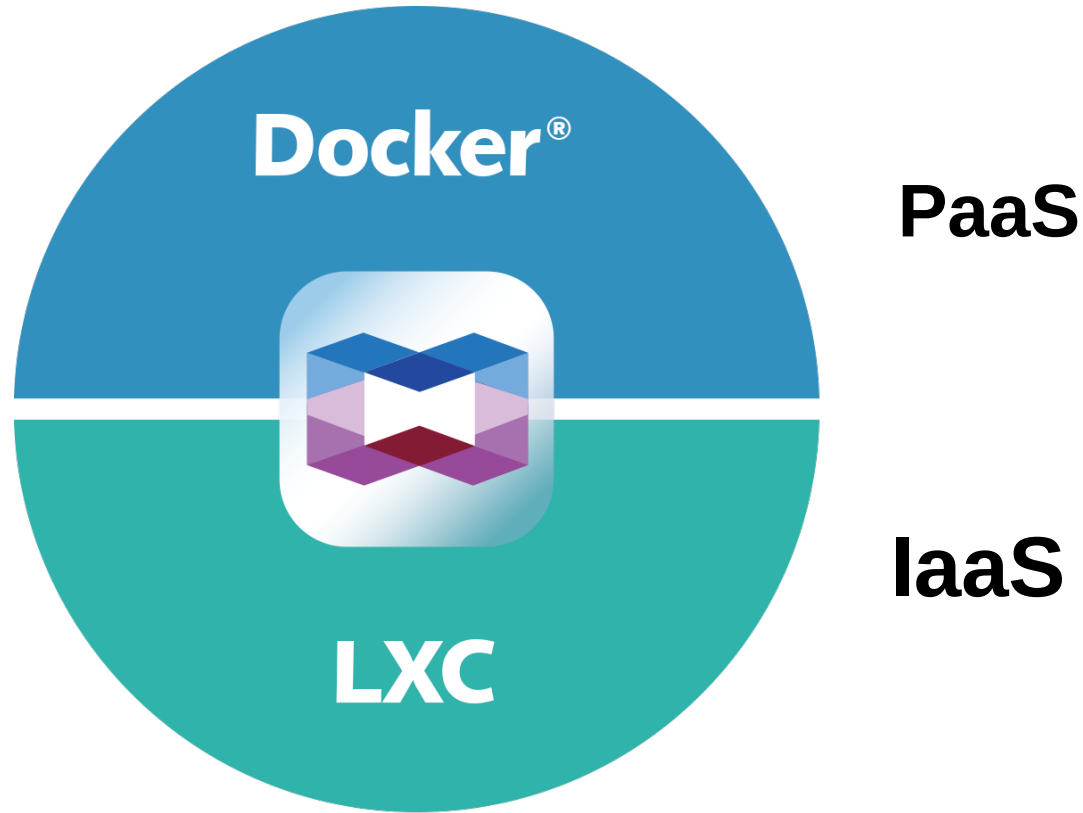




# Insight :: Containers



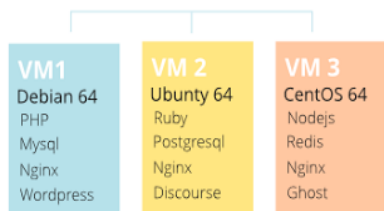
# Insight :: Containers



# Insight :: LXC v. Docker

## LXC

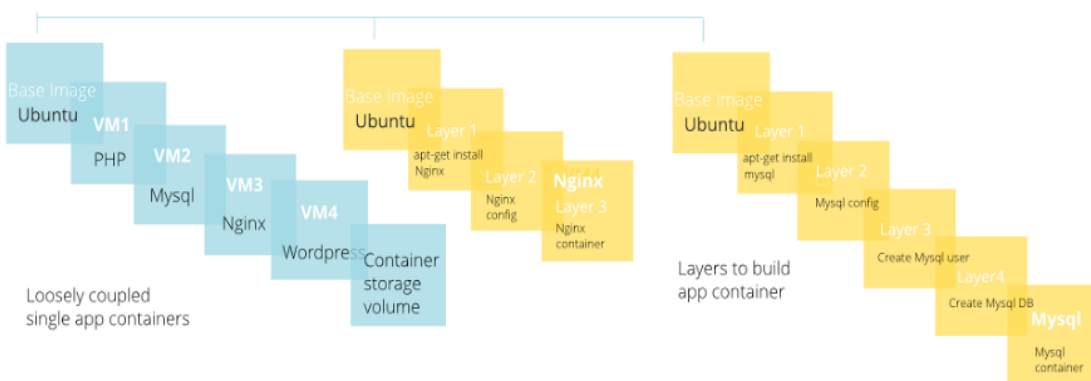
### Host



- Filesystem neutral
- Containers are like VMs with a fully functional OS
- Data can be saved in a container or outside
- Build loosely coupled or composite stacks

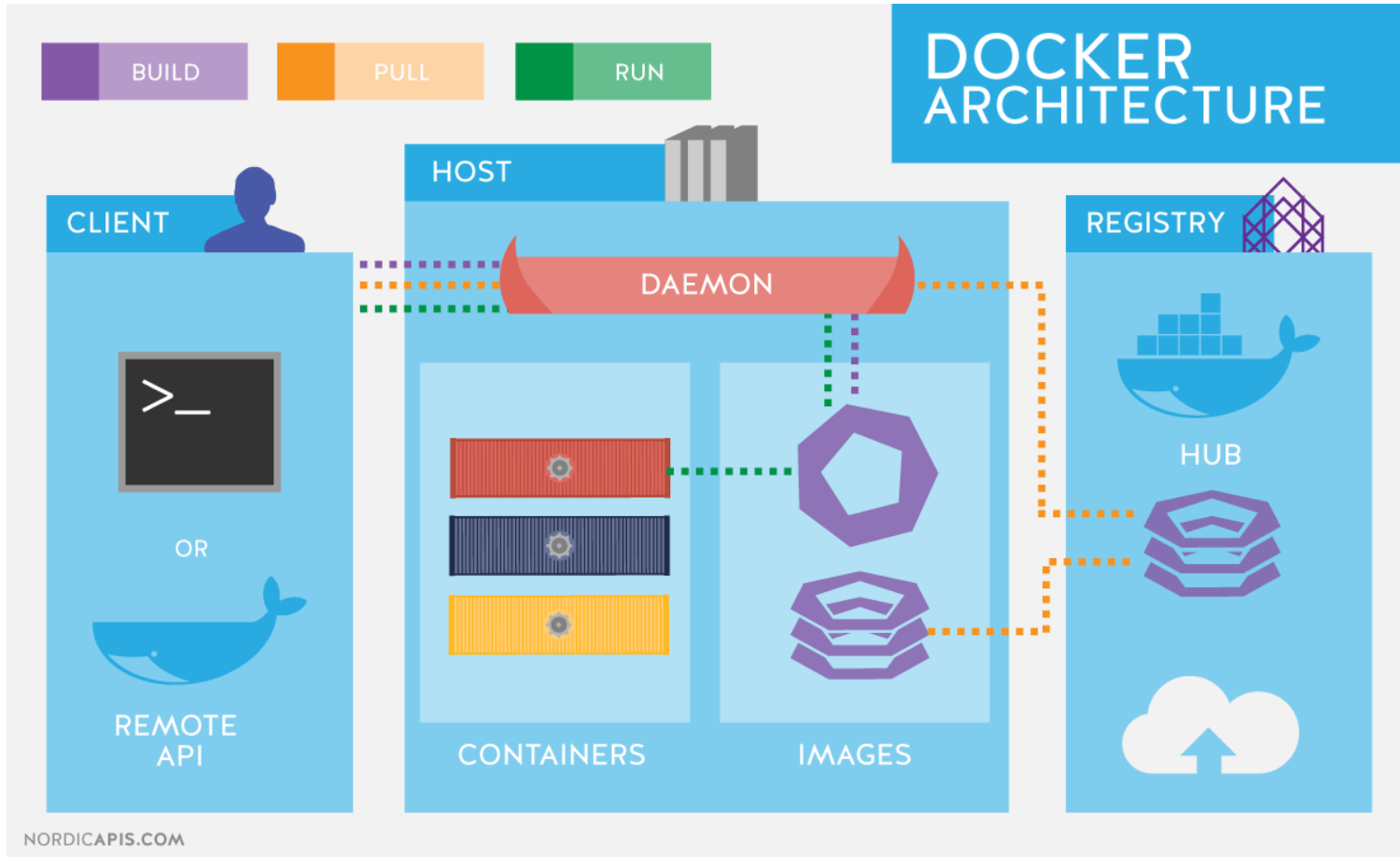
## Docker

### Host



- Containers are made up of read only layers via AUFS/Devicemapper
- Containers are designed to support a single application.
- Instances are ephemeral, persistent data is stored in bind mounts to host or data volume containers

# Insight :: Containers



# Insight :: NetFlow Alternative

Elastic Stack (X-Pack)



logstash



elastic



---

**Replacement**

# Insight :: Network Security & Visibility Delivery Platform



Stealthwatch Cloud

[https://www.youtube.com/watch?v=MZr\\_V2L-RXk](https://www.youtube.com/watch?v=MZr_V2L-RXk)



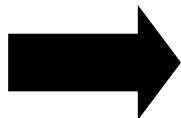
<https://www.youtube.com/watch?v=sPRe6xsgXQg>



elastic

<https://www.elastic.co/guide/en/logstash/current/netflow-module.html>

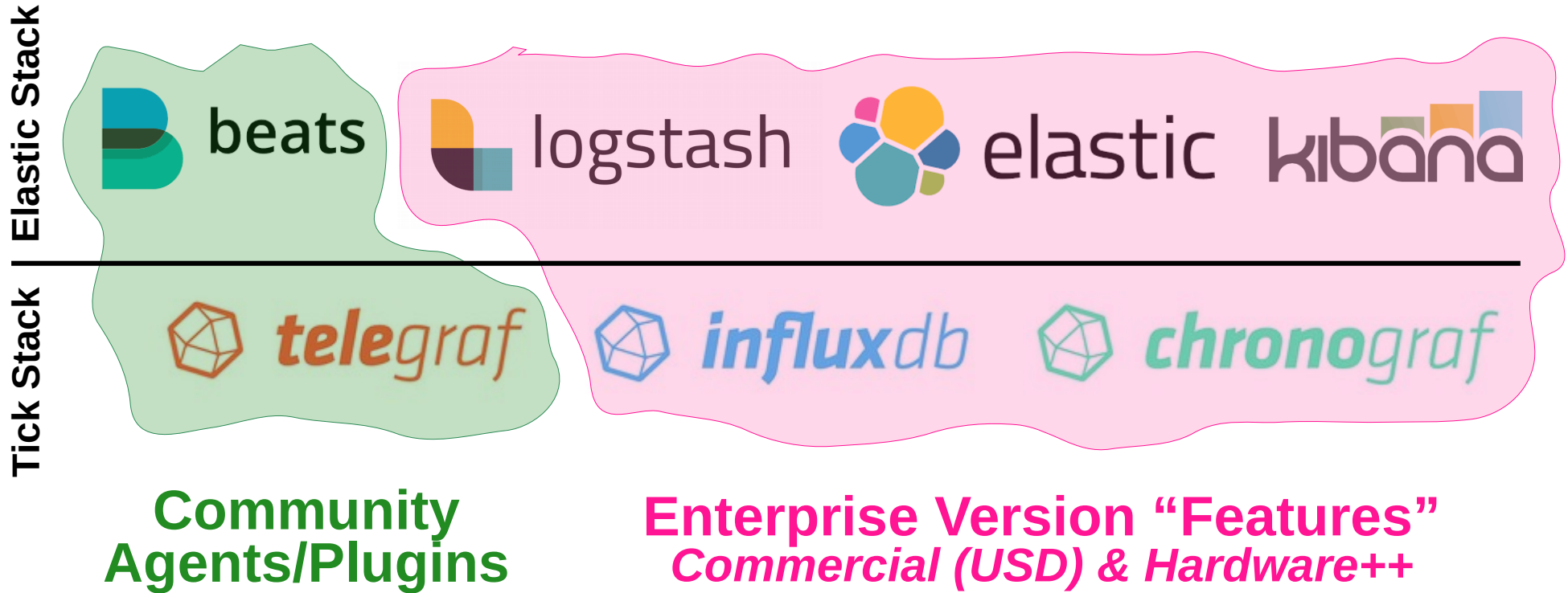
**Cisco NetFlow, S/C/J/  
O/Q-flow, IPFix Based**



**Layer 7, RTT, SrvRT,  
Top\*, L2/L3 Match**

**Agentless  
Real Time  
Flow Based  
Flow Collector  
Concentrator  
Buffer Based  
Packet Analyzer  
Flow Forwarding**

# Insight :: Alternate Agent Stack



# Insight :: Cloud Monitoring



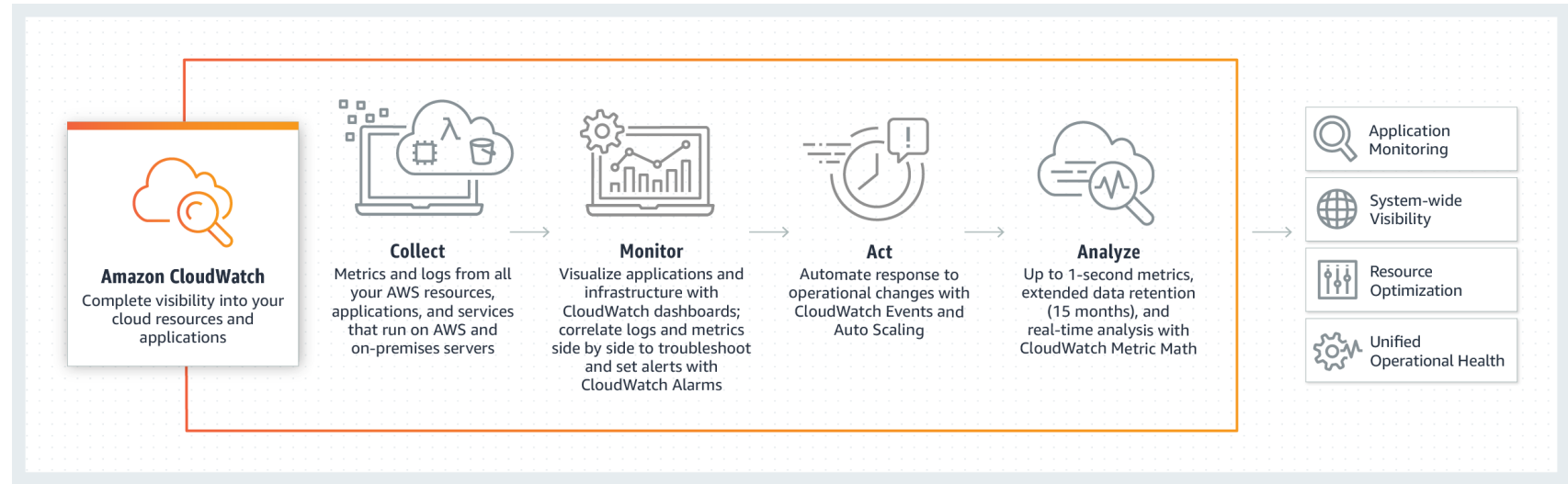
**Amazon CloudWatch**



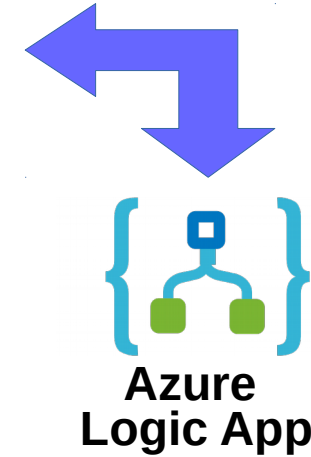
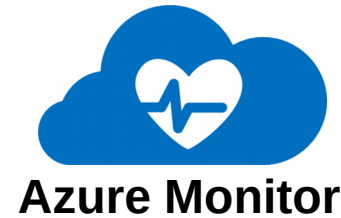
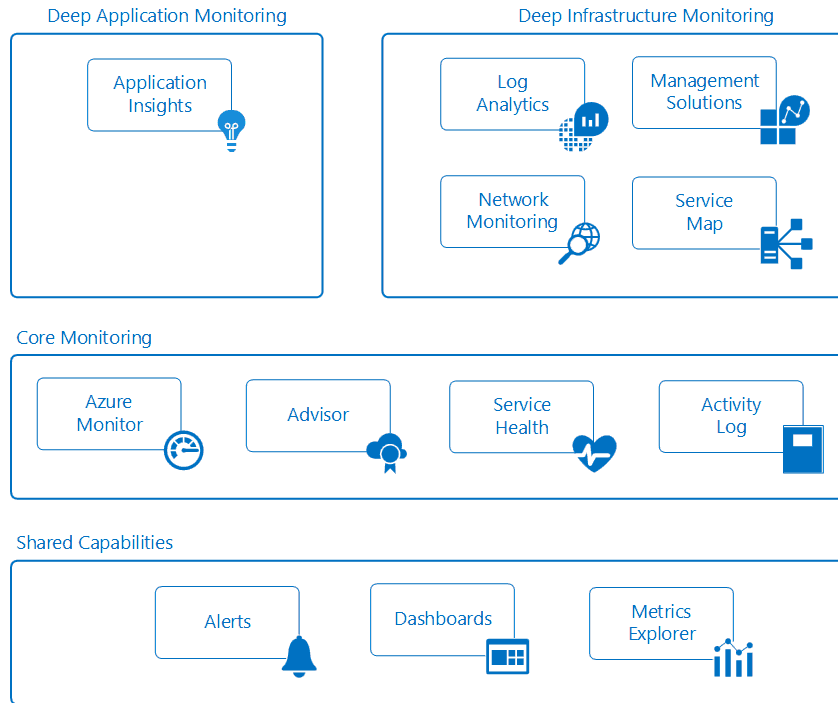
**Azure Monitor**



# Insight :: Cloud Monitoring

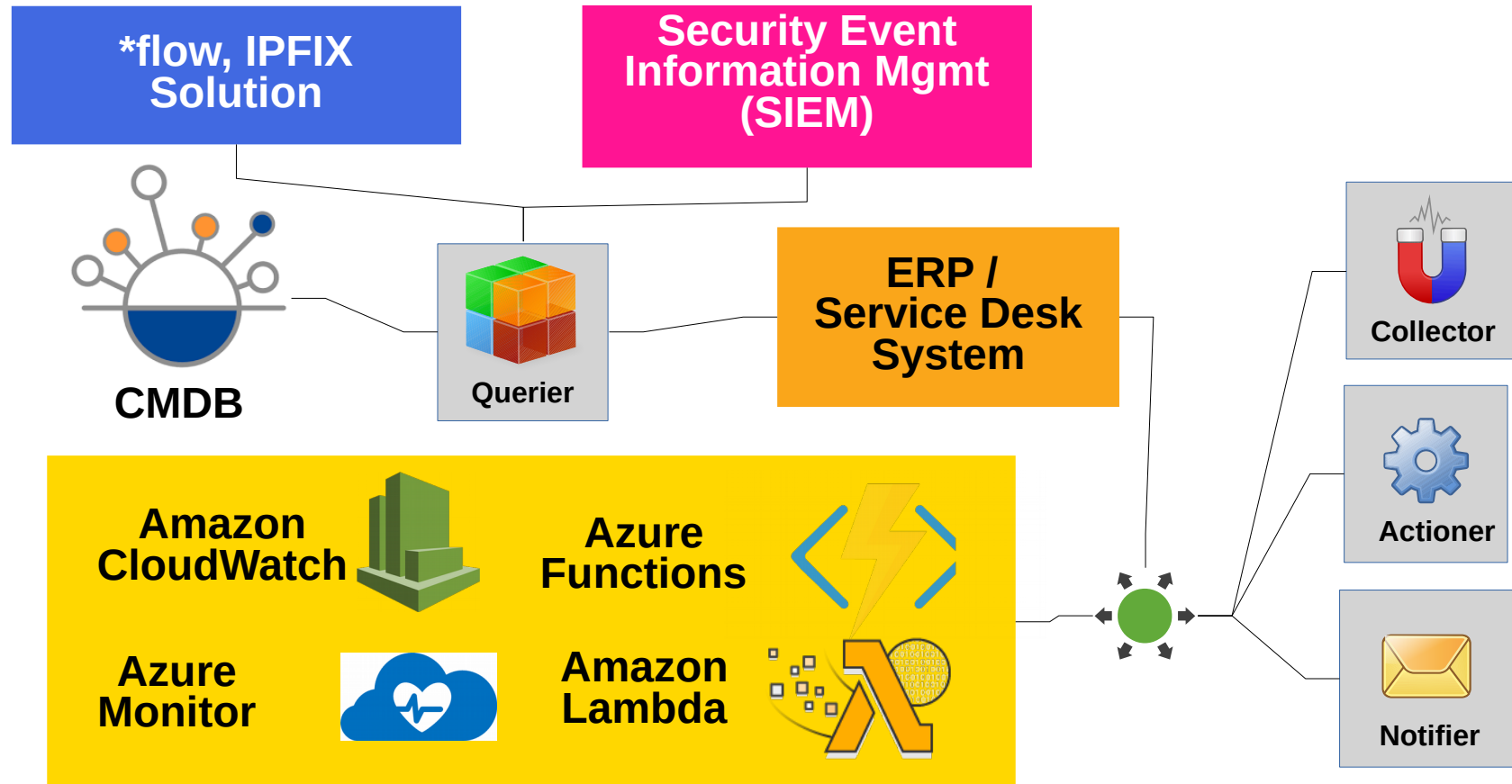


# Insight :: Cloud Monitoring

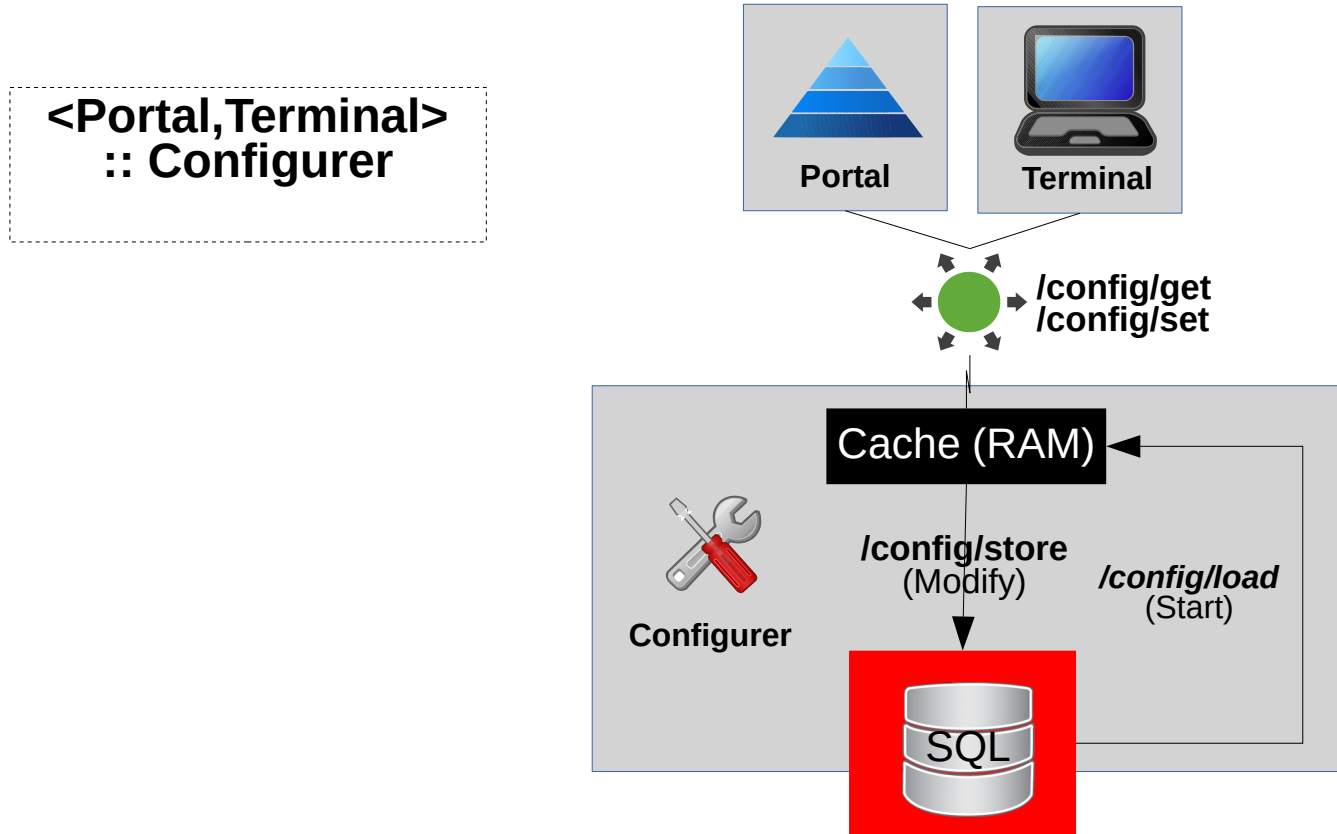


<https://docs.microsoft.com/en-us/azure/monitoring-and-diagnostics/monitoring-action-groups-logic-app>  
<https://docs.microsoft.com/en-us/azure/connectors/connectors-native-http>

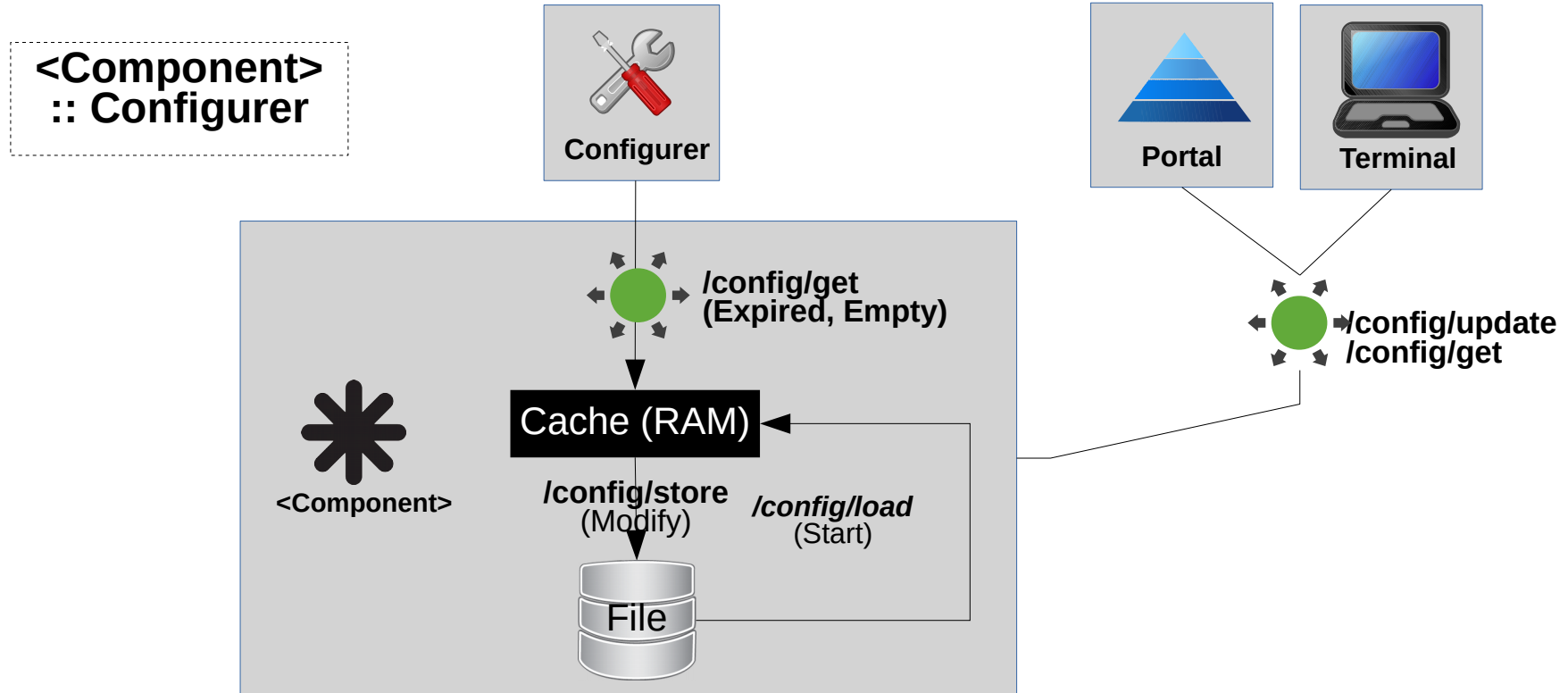
# Diagrama de Interfaces



# Insight :: Colaboration

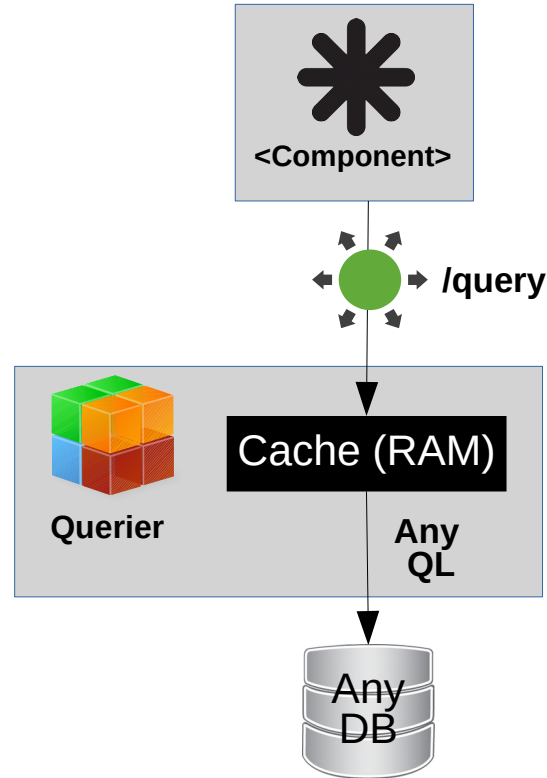


# Insight :: Colaboration



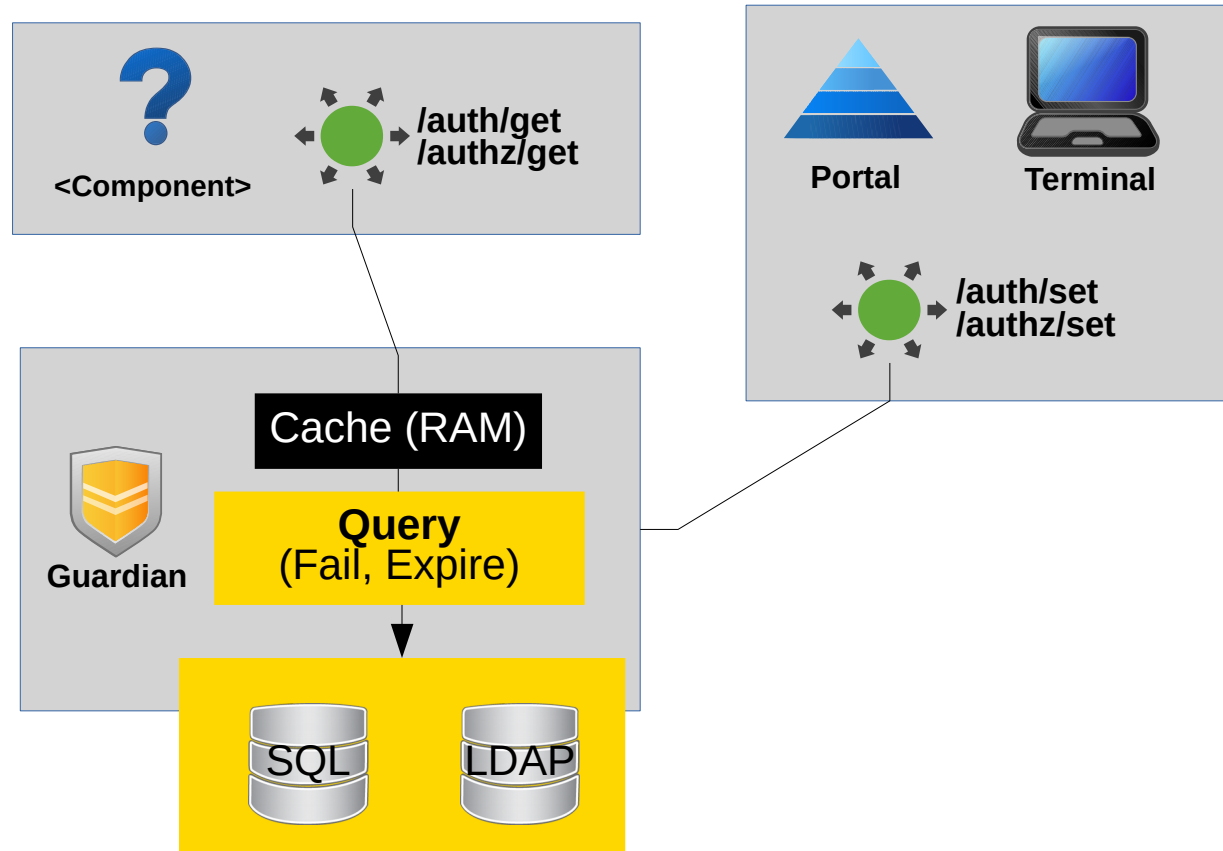
# Insight :: Colaboration

**<Component>  
:: Querier**

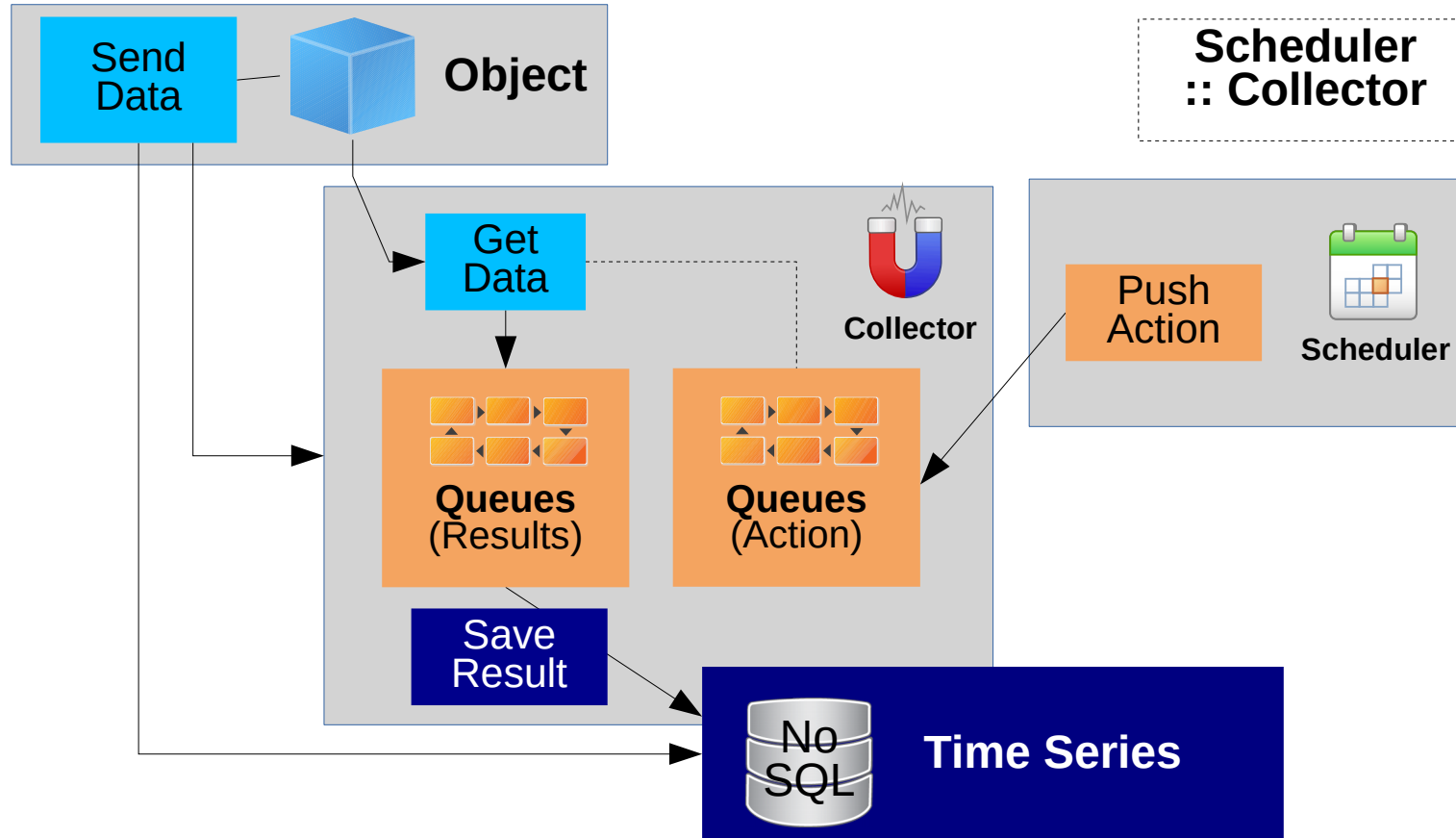


# Insight :: Colaboration

**<Component>  
:: Guardian**

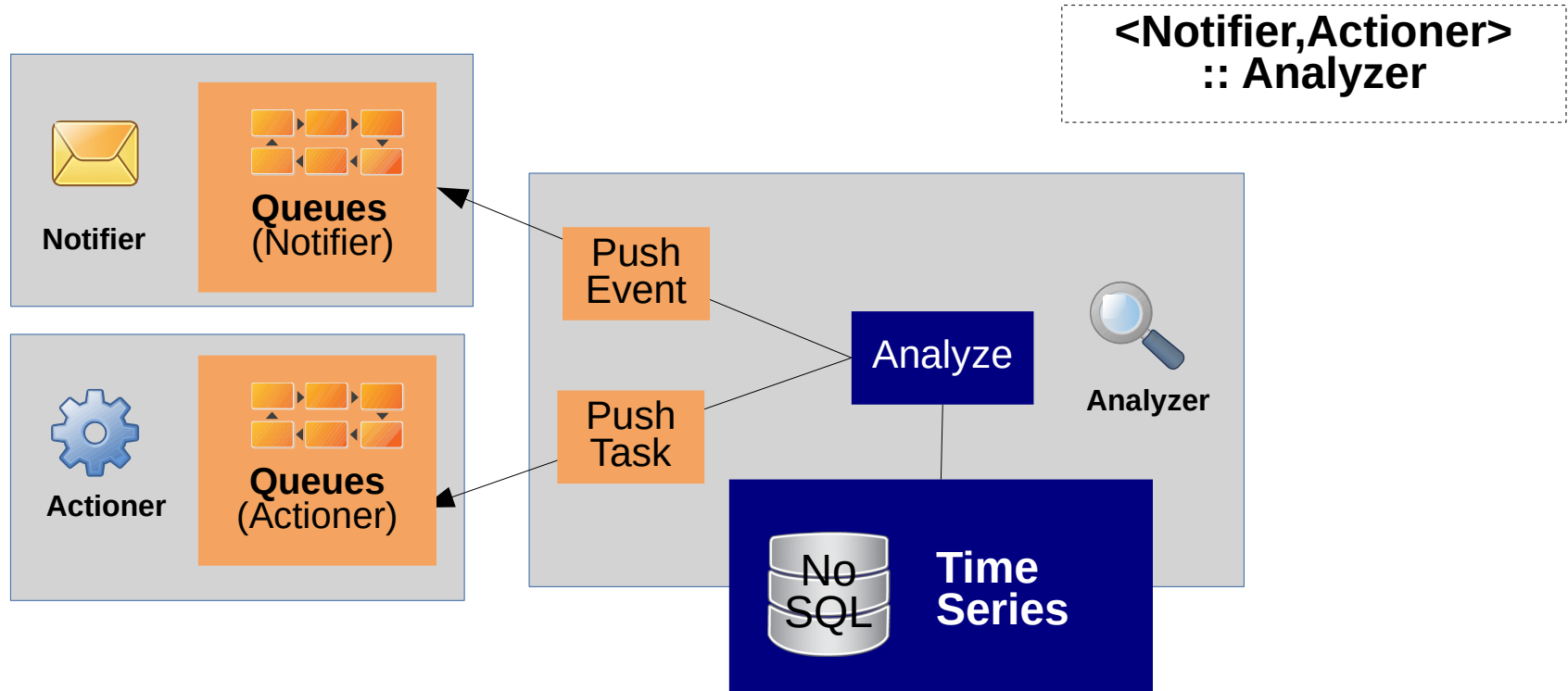


# Insight :: Colaboration





# Insight :: Colaboration



# Insight :: Colaboration

**<Portal,Terminal>  
:: Grapher**

